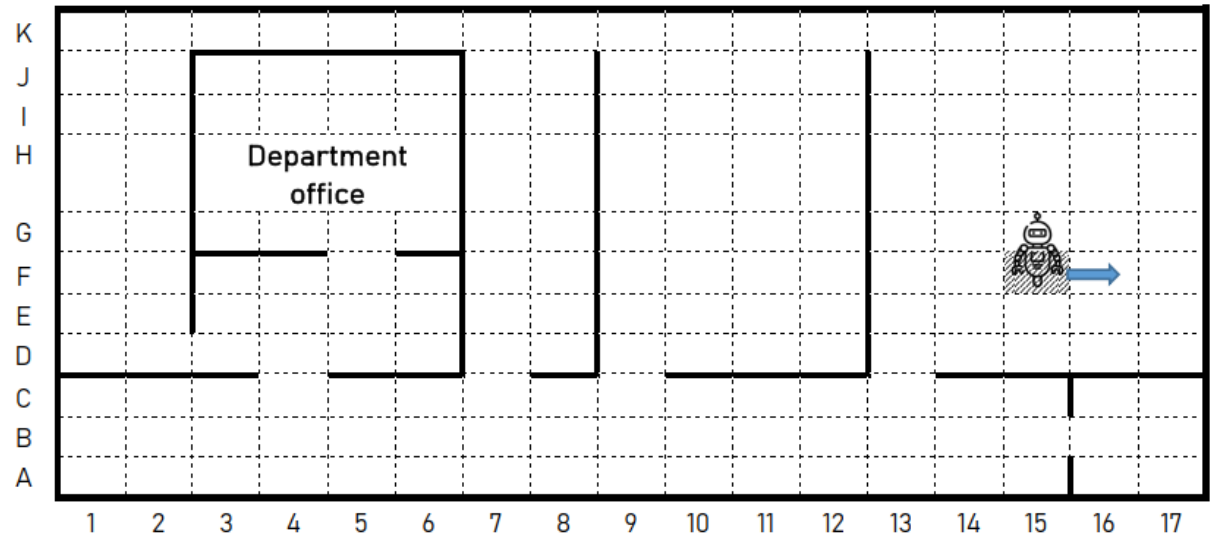
Exercise 01: We consider that a robot can move in a specific framework, a grid: Each box corresponds to a possible position of the robot.



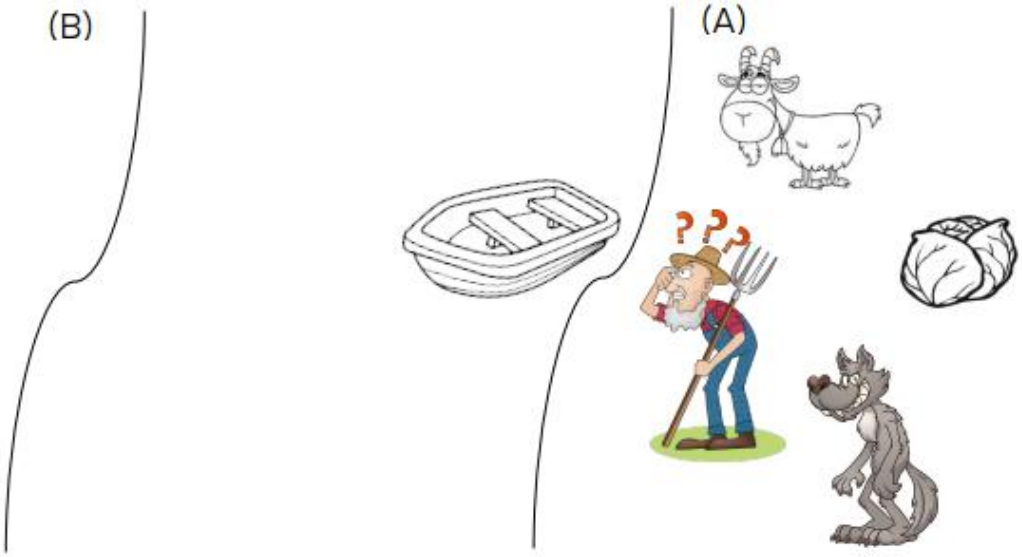
Let's define the following orders that the robot can execute:
- < **forward** >: the robot moves a single space forward in the direction of the guiding arrow.
- < **rotate** >: the robot turns 90° to the right (clockwise).
- < **position** >: the robot is placed at the state "Ready to go" (can be used once at the deployment of the robot).

By using the previous orders:
a. Give the sequence of orders to be executed in order to move the robot from its initial position to the department office.
b. If we assume that the robot can also execute the order < **repeat** > defined by:

repeat N times (where N>0)
    <Sequence of orders>
end-of-repeat

Answer the previous question using the order **repeat**.

Exercise 02: A farmer has to cross a river from the bank (A) to the bank (B), with his purchases: the wolf, the goat, or the cabbage (as depicted in the figure below)

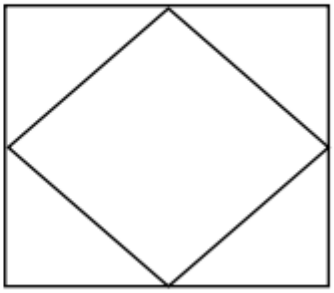


Knowing that on the boat, the farmer could carry only himself and a single one of his purchases: the wolf, the goat, or the cabbage, and using the following actions:

1. < **Cross(Obj,A,B)** >: farmer crosses from (A) to (B) with the element **Obj.**
2. < **Cross(Obj,B,A)** >: farmer crosses from (B) to (A) with the element **Obj.**
3. < **drop_off(Obj)** >: farmer deposits the element **Obj** (from boat to the bank).
4. < **Back** >: farmer crosses the river alone.

If left unattended together, the wolf would eat the goat, or the goat would eat the cabbage. The farmer's challenge is to carry himself and his purchases to the other bank of the river, leaving each purchase intact. Give a solution to the former problem.

Exercise 03: We would like to draw the following pattern



Respecting the following constraints:
l We start from a given point (x, y);
l You never go back through an already traced line;
l We draw the whole pattern without ever lifting the pencil.
We assume that the following instructions are known:
l **place(x, y):** places the pencil at the point of coordinates (x, y);
l **plot(x, y):** joins the current point to the point of coordinates (x, y);
l **finish():** ends the processing.
1. Write the sequence of instructions that allows us to draw the given pattern (here above).

Solutions:

Exo 1

a) Sequence without loops: `<rotate>` = 90° clockwise

```
position
rotate
rotate
forward
forward
rotate
rotate
rotate
forward
forward
forward
rotate
forward
forward
forward
forward
forward
forward
forward
forward
forward
rotate
forward
forward
forward
rotate
forward
rotate
rotate
rotate
forward
```

b) Sequence using `repeat`:

```
position
repeat 2 times
    rotate
end-of-repeat
repeat 2 times
    forward
end-of-repeat
… etc.
```

Exo 2

- **Items:** Wolf (W), Goat (G), Cabbage (C).
- **Constraints:** Wolf cannot be left alone with Goat; Goat cannot be left alone with Cabbage.
- **Available actions:** `Cross(Obj, A, B)`, `Cross(Obj, B, A)`, `drop_off(Obj)`, `Back`.

Plaintext

```
1. Cross(Goat, A, B)
2. drop_off(Goat)
3. Back
4. Cross(Wolf, A, B)
5. drop_off(Wolf)
6. Cross(Goat, B, A)
7. drop_off(Goat)
8. Cross(Cabbage, A, B)
9. drop_off(Cabbage)
10. Back
11. Cross(Goat, A, B)
12. drop_off(Goat)
```

(Alternatively, cabbage and wolf can be swapped in steps 4 and 8).

Exo 3

```
place(0, 0)
plot(2, 0)
plot(4, 0)
plot(4, 2)
plot(4, 4)
plot(2, 4)
plot(0, 4)
plot(0, 2)
plot(0, 0)
plot(0, 2)
plot(2, 4)
plot(4, 2)
plot(2, 0)
plot(0, 2)
finish()
```